Geographic and Linguistic Bias in LLMs for Air Pollution Policy: A 25-Country, 14-Model, 4-Language, 5-Task Benchmark

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Abstract

Public environmental managers in the Global South increasingly ask large language models (LLMs) about binding air-quality standards, measured concentrations, and health evidence — a domain where a wrong answer can shape a regulatory brief or an emergency response. Whether these systems serve the Global South as reliably as the Global North, and whether prompt framing changes the answer, has not been measured for an applied policy domain with verifiable ground truth. $PM_{2.5}$ is the leading environmental risk factor for premature mortality, and its burden falls disproportionately on the Global South [1–4]. We benchmark **25 countries, 14 LLMs, four languages, and five task types** (normative recall, local measured data, health-evidence synthesis, policy instruments, and applied recommendation) in air pollution policy, against official ground truth, with a within-subject persona factor (neutral versus public environmental manager) and a within-country English/native-language contrast. The design was pre-registered for 15 countries; we then extended it post hoc to 25 (seven more Global North, three more Global South) to gain power, reporting every effect alongside its pre-registered 15-country value.

From LLM-as-judge composite scores we find a Global North/South **tier** gap of +6.2 pp (bootstrap 95% CI $[+3.7, +8.6]$). A development gradient appears in the 25-country sample (Spearman $\rho = 0.51$ with HDI, $p = 0.004$; Mann-Kendall $p = 0.018$; leave-one-out robust). It was undetectable at the pre-registered $n = 15$ ($\rho = 0.14$) and sits below our pre-registered 0.55 effect-size threshold, and it tracks development, not linguistic-resource class (Joshi null). Three harms are robust. Native-language prompting *lowers* accuracy (-2.1 pp overall, $p = 2 \times 10^{-3}$; -7.8 pp for Hindi), so the local language does not help and can hurt. Accuracy collapses on technical-standard and local-datum recall (T1, T2 ≈ 0.37 vs 0.64 elsewhere; $\delta = -0.64$), a regulatory-cutoff floor. And the Brazilian-Portuguese regional model is the weakest of all ($\delta = -0.51$). Three intuitive fixes therefore fail together: neither a local-manager persona (permutation $p = 0.26$), nor the local language, nor a regional model closes the gap. On mechanism, the pre-registered corpus proxy (Wikipedia size) is null; a country-coverage proxy (Wikidata sitelinks) correlates with accuracy ($\rho = 0.54$, $p = 0.005$, surviving multiplicity correction) but attenuates to non-significance after adjusting for development ($p = 0.13$), so we report country coverage as a *suggestive, exploratory* channel — better than language-corpus size, as expected for a gap measured in English — not as an established mechanism. Reliability across a four-vendor judge panel is strong (ICC of the panel mean = 0.89), and ground truth is anchored to official primary sources for all 25 countries; the study uses one panel-validated primary judge and a documentary ground-truth audit without a human-gold layer. Protocol, prompts, and analysis code will be released on publication.

Keywords: large language models; geographic bias; Global South; epistemic justice; policy research; benchmark; multilingual NLP

1 Introduction

Large language models (LLMs) have moved from research curiosity to working infrastructure of public administration in under five years: an OECD survey of core government functions documents generative-AI adoption across policymaking and public-service delivery, including deployments that help public-administration staff search regulations and draft technical summaries [5]. Environmental agencies, municipal secretariats, and applied policy analysts increasingly draft technical syntheses, translate regulatory norms, locate official monitoring data, and characterize institutional frameworks with such assistance. Air pollution policy is a domain where this adoption carries direct stakes for public health. Fine particulate matter ($PM_{2.5}$) is the leading environmental risk factor for premature mortality worldwide [1, 2], and the burden falls disproportionately on the Global South [1]. When a public environmental manager in São Paulo, Lagos, or Jakarta asks an LLM about an ambient air quality standard, an attainment deadline, or the epidemiological evidence behind an intervention, an inaccurate answer is not a benign error. It can propagate into a regulatory brief, a procurement decision, or an emergency response during a pollution episode.

The question this paper addresses is whether LLMs serve Global South air pollution policy as reliably as they serve the Global North, and whether the way a query is framed changes the answer. Two recent strands of evidence make this question urgent rather than hypothetical.

Geographic factual bias is real and measurable. Moayeri et al. [6] documented 1.5× higher factual error rates for Sub-Saharan African countries versus North American countries across 20 models and 11 World Bank indicators. Manvi et al. [7] showed that LLM zero-shot ratings on subjective topics correlate strongly with income level (ρ up to 0.70 against socioeconomic conditions). Mirza et al. [8] found that GPT-4 systematically privileges statements about the Global North over the Global South in factual verification tasks. These results establish the phenomenon, but none of them targets a single high-stakes applied domain with verifiable official ground truth, and none asks whether the bias can be modulated by prompt framing.

Linguistic and corpus asymmetry is structural. Joshi et al. [3] mapped a six-class hierarchy of linguistic resource availability in NLP, showing that the languages of most Global South countries sit in the under-served lower classes. Under-representation in the training corpus is the dominant explanation for downstream geographic bias [9]. This asymmetry is not incidental; it reproduces longstanding patterns of uneven digital and epistemic infrastructure that decolonial scholarship has analysed for AI systems [4].

We identify three gaps that current work leaves open for a domain like air pollution policy.

Gap 1: no benchmark targets air pollution policy with official ground truth. Existing evaluations use numeric World Bank recall (WorldBench), subjective ratings (Manvi et al.), journalistic fact-checking (Global-Liar), or everyday cultural knowledge (BLEnD [10]). None measures the recall and synthesis tasks a public environmental manager actually delegates to an LLM: stating a binding ambient standard, retrieving an officially measured concentration, summarizing the health-burden evidence, identifying the legal instruments and executing agencies, or recommending an operational response. These tasks anchor to authoritative sources (national environmental councils, state monitoring agencies, the World Health Organization, the Global Burden of Disease estimates), which makes accuracy verifiable in a way that trivia benchmarks are not.

Gap 2: persona framing is unexamined as an experimental lever. Practitioners routinely prefix queries with a role ("As a municipal environmental secretary, ..."), and persona or role conditioning is a documented prompt pattern in deployed prompting workflows [11]. Whether adopting the persona of a public environmental manager *reduces* the Global South accuracy gap (by steering the model toward policy-relevant, regulation-compliant responses) or *amplifies* it (by inviting confident fabrication of regulatory specifics the model never learned) has not been tested in a pre-specified design. We treat persona as a within-subject experimental factor rather than an uncontrolled confound.

Gap 3: the corpus-representation mechanism is asserted, not tested for this domain. The link between corpus representation and accuracy is assumed but rarely measured against confounders, and rarely separated into its two channels. We distinguish *language-corpus* size (how much text exists in a country's language: mC4 token volume [12], OSCAR size [13], Joshi class) from *country-coverage* breadth (how broadly the corpus is about a country: Wikipedia and Wikidata coverage), and test which predicts air pollution policy accuracy after adjusting for Human Development Index — a distinction that matters because the geographic gap is measured in English.

Contributions of this work

We present a benchmark designed for these gaps, with hypotheses, sampling, and analysis plan pre-specified before data collection [14], executed across 15 countries and then extended to 25. Our contributions are:

1. **An air pollution policy task suite with verifiable ground truth.** We construct five task types mapped to real decision activities of a public environmental manager: recall of binding ambient standards and emission limits (T1), retrieval of officially measured

local concentrations (T2), synthesis of epidemiological health-burden evidence (T3), identification of policy instruments and executing agencies (T4), and rubric-scored applied recommendation for an operational scenario (T5). Each task anchors to official national or international sources, and every reference in the ground truth is verified against a resolvable DOI or URL before entering the dataset.

2. **Persona prompting as a pre-specified experimental factor.** We manipulate persona within subject (neutral query vs. public environmental manager) and test, as a co-primary confirmatory hypothesis (H6), whether the persona condition reduces the Global South accuracy gap. We pre-specify both the reduction direction (Norm-Compliance hypothesis) and the amplification direction (Persona-Vacuum hypothesis), reporting whichever the data support.

3. **Theoretically-triangulated country sampling.** We select 15 countries stratified along independent axes: UNCTAD North/South classification (operationalising coloniality [4]), the Joshi et al. six-class linguistic-resource taxonomy [3], and Human Development and income indicators, later extended to 25 to gain power for the gradient and mechanism tests. This separates geographic, linguistic, and economic contributions to the bias, and shows — as a methodological by-product — that the development gradient is underpowered at $n = 15$ but significant at $n = 25$.

4. **Mechanistic test separating the two corpus channels.** We test the corpus-representation hypothesis (H4) using both language-corpus and country-coverage proxies, with partial-correlation controls for HDI. The pre-registered language-size proxy is null; a country-coverage proxy is a suggestive but development-entangled lead, which we report as exploratory rather than as an established mechanism.

5. **Regional-model probe.** We include a Brazilian-Portuguese instruction-tuned open-weight model (Cabra-Mistral 7B v3) to test, as a secondary confirmatory hypothesis (H3), whether regional tuning narrows the Portuguese-language gap for Brazil on air pollution policy tasks.

6. **Open resources.** The full protocol, prompts (after validation), ground-truth source documents, analysis code, and figures will be released on publication under CC-BY-4.0 and MIT licenses.

Theoretical framing

We interpret our findings within the *coloniality of knowledge* framework [4, 15] and its data and infrastructural articulations [16, 17]. LLMs are compressed statistical summaries of corpora produced under conditions of uneven digital infrastructure and uneven publication access. The languages and regions that Joshi et al. place in the under-served classes are precisely those where air pollution kills the most people [1, 3]. When a Global South environmental manager consults a model trained largely on Global North text to govern a local air quality problem, the resulting decision risks inheriting the asymmetries of the corpus. This positions our empirical work as a test of one link in a theorised reproduction cycle (training-corpus asymmetry $\rightarrow$ LLM knowledge asymmetry → policy decision asymmetry), and the persona manipulation as a test of whether prompt framing can interrupt that link.

The paper proceeds as follows. Section 2 positions our work relative to the literature. Section 3 presents the pre-specified design. Section 4 reports the empirical findings. Section 5 interprets them through both statistical and critical lenses and considers practical implications for air pollution governance. Section 6 closes.

2 Related Work

We organize prior work along four threads: the discovery and measurement of geographic factual bias, multilingual and cultural extensions, alignment and mitigation attempts, and LLM use in policy and health decision contexts. For each we state how the air pollution policy focus and the persona manipulation of the present study extend the thread.

2.1 Geographic factual bias

Manvi et al. [7] established that frontier LLMs carry systematic biases against regions of lower socioeconomic conditions, with Spearman correlations up to 0.70 between model-generated ratings of attractiveness, morality, and intelligence and per-capita income. The outcome space (subjective ratings) leaves open whether the same bias appears in objective recall of regulatory and epidemiological facts, which is what air pollution policy requires.

Moayeri et al. [6] introduced WorldBench, quantifying factual recall against World Bank indicators across 20 LLMs and 11 indicators and documenting 1.5× higher absolute relative error for Sub-Saharan Africa versus North America. WorldBench also pioneered automatic detection of citation hallucination, where models cite the World Bank while producing false statistics. The design is bounded by the macro-economic indicator space; an environmental manager asks narrower, source-specific questions (a national ambient standard, an attainment phase deadline, a city-year measured concentration) whose ground truth lives in environmental councils and monitoring agencies rather than in a single aggregated database.

Mirza et al. [8] (Global-Liar) extended the question to temporal stability, showing that newer GPT-4 versions do not monotonically improve and that the privileging of Global North statements persists across versions. Their contribution is temporal and within-model rather than cross-model, cross-language, or domain-specific.

These studies establish that the phenomenon exists and is measurable. None isolates a single applied domain with verifiable official ground truth, and none treats prompt framing as a manipulable factor.

2.2 Multilingual and cultural extensions

Myung et al. [10] (BLEnD) introduced question-answer pairs covering 16 countries and 13 languages in everyday cultural knowledge. Their finding that LLMs perform better in the native languages of high-resource cultures but worse in those of low-resource ones motivates our exploratory language hypothesis (H2) and connects directly to the resource hierarchy of Joshi et al. [3]. Yet BLEnD measures cultural-knowledge accuracy, not the accuracy of regulatory and health-policy recall, and its country selection is not stratified theoretically.

Regional benchmarks proliferated: BRoverbs [18] for Brazilian proverbs, TiEBe [19] for regional historical events, and ALBA [20] and AMALIA [21] for European Portuguese. These document concrete failures in culturally grounded tasks but do not address applied policy use, and each is limited to one or two languages. Work on Portuguese corpus construction [22] underscores how Common Crawl snapshots under-represent Lusophone content, the corpus asymmetry our H4 tests.

2.3 Alignment and mitigation

Opuszko and Böhm [23] investigated whether reasoning models reduce geographic bias, finding that reasoning helps in aggregate but does not eliminate regional disparities. Kerche et al. [24] proposed a place-based typology of LLM biases, situating geographic bias within a broader sociotechnical landscape. These contributions are conceptual and aggregate; we add a domain-specific, pre-specified empirical test and introduce persona as a candidate mitigation lever rather than a fixed property of the model.

2.4 LLMs in policy and health decision contexts

A parallel literature examines LLM use in specific policy settings. He et al. [25] evaluated LLM policy recommendations for homelessness across four cities (including Barcelona, Johannesburg, and South Bend), finding "context-blind rigidity" in which models apply stable internal heuristics poorly calibrated to local realities and align more closely with Global North experts. Their result shows the bias matters for policy in a single domain and a single task type (recommendation). We generalize to a regulatory and health domain with five task types spanning recall, synthesis, and recommendation, and we add the persona manipulation absent from their design.

In health specifically, Kozlakidis et al. [26] argue that LLM hallucinations carry acute risk in regulatory environments with limited technical capacity, noting that language and cultural mismatch between globally trained models and local practice increases error. Air pollution policy sits at the intersection of this risk and the mortality burden documented by the Global Burden of Disease estimates [1], which makes accurate retrieval of standards and health evidence a public-health concern rather than a benchmarking abstraction.

2.5 Persona and role conditioning

Role or persona conditioning is a documented prompt pattern in deployed prompting [11], and a growing body of work reports that assigning an expert role can shift model outputs in either direction depending on task and domain. Systematic evidence is sobering: across four LLM families and 2,410 factual questions, adding expert personas to system prompts did not improve accuracy over a no-persona control, and across 162 roles it slightly *reduced* accuracy on average [27]; an independent replication likewise finds that expert personas do not improve factual accuracy [28]. Whether persona conditioning helps or harms factual reliability in high-stakes, knowledge-intensive domains is therefore unsettled and, to our knowledge, untested for geographic equity. We therefore treat persona as a pre-specified experimental factor and test both the Norm-Compliance hypothesis (persona reduces the gap) and the Persona-Vacuum hypothesis (persona amplifies it).

2.6 Positioning of the present work

Our design differs from prior work in four concrete ways: (i) a single high-stakes applied domain (air pollution policy) with verifiable official ground truth, rather than numeric trivia or cultural knowledge; (ii) persona as a pre-specified within-subject experimental factor, rather than an uncontrolled framing choice; (iii) theory-driven country sampling on independent axes (UNCTAD coloniality, Joshi linguistic resource, development), rather than regional convenience; and (iv) an explicit corpus-representation mechanism test with sensitivity analysis. No prior study combines a regulated public-health domain, a persona manipulation, and a pre-specified mechanism test.

3 Methods

3.1 Research design

We specify a pre-specified, factorial, multi-country benchmark experiment to quantify geographic and persona-modulated bias in 14 large language models on air pollution policy tasks relevant to public environmental management. The pre-registered design crosses 15 countries (stratified along three theoretically grounded axes), 14 LLMs spanning five access tiers, one theory-driven domain (air pollution policy), 5 task types, and 2 persona conditions (neutral versus public environmental manager), with 2 stochastic replications per prompt-model-persona cell; a post-registration extension (Section 3.3) enlarges the sample to 25 countries. The stimulus set crosses each country's five tasks and two personas in English, plus a within-country native-language rendering for the nine countries with a target native language, scored on the $[0, 1]$ composite. The study protocol fixes, in advance of data collection, the hypotheses, primary models, persona protocol, exclusion criteria, and multiple-comparison corrections. A Brazil-only collection executed under an earlier three-domain design is reclassified as an extended pilot and is reported separately; it informed prompt calibration but contributes no confirmatory observations.

3.2 Country selection

Countries were selected via theoretical sampling [29] triangulated across three independent classifications. First, the **UNCTAD Global North/South classification** [30] operationalizes the coloniality-of-knowledge framework that motivates this study [4]. Second, the **Joshi et al. (2020) six-class taxonomy** of linguistic-resource representation in NLP corpora provided an orthogonal stratification dimension [3]. Third, **World Bank income groups (FY26)** [31] captured economic position independently of the previous two axes.

The final sample of 15 countries — Brazil, Mexico, Argentina, Peru (Latin America); Nigeria, South Africa, Kenya, Egypt (Africa); India, Indonesia, Bangladesh, the Philippines (Asia); and the United States, Germany, Japan (Global North reference) — covers four world regions, five Joshi classes (1 through 5), and four World Bank income groups. The twelve Global South countries also vary in domestic air quality governance maturity, which is relevant to ground-truth availability (Section 3.9). Countries with compromised statistical governance (e.g., active conflict) or without a publicly retrievable national ambient air quality standard were excluded; Oceania and Joshi Class 0 languages were not represented given feasibility constraints and are acknowledged as scope limitations.

3.3 Confirmatory sample expansion (post-registration)

After the pre-registered 15-country analysis, and *transparently reported here as a post-registration extension rather than part of the locked protocol*, the sample was enlarged to **25 countries** to increase statistical power for the gradient (H1) and corpus-mechanism (H4) tests and to multiply the within-country language pairs available for H2. Ten countries were added under the identical prompt-construction, ground-truth, collection, and scoring procedures. Six are **Global North** references chosen purely to broaden the high-resource end of the gradient — the United Kingdom, Canada, Australia, South Korea, France, and Italy. Four are **native-language-pair** countries that add Spanish (Colombia, Chile) and Portuguese (Portugal, Angola) contrasts for H2; of these, Portugal is also Global North while Colombia, Chile, and Angola are Global South. The two roles overlap, so by tier the extension adds seven Global North countries (the six references plus Portugal), raising the Global North cell from three to ten, and three Global South, while by language it raises Spanish-speaking pairs from three to five and Portuguese-speaking pairs from one to three. Because this extension was decided after observing the initial results, every effect computed on the 25-country sample is reported alongside its pre-registered 15-country counterpart, and no pre-registered conclusion is revised on the basis of the extension alone; the expansion is treated as a power-and-robustness check, not as a new confirmatory test. Angola, like Nigeria and Argentina, has *no* national ambient $PM_{2.5}$ standard, providing additional no-standard cases in which a faithful model must decline to state a non-existent threshold rather than fabricate one.

To multiply H2 pairs without adding new languages, two additional question variants per (country, task) were also generated for the five original native-pair countries via a high-quality generator model and rendered in both English and the native language; these variants are used *only* for the paired English-versus-native contrast (H2), in which any imperfection in the generated ground truth cancels between the two members of a pair, and are excluded from absolute-accuracy comparisons (H1, H3).

3.4 Country covariates

For each country we compiled developmental and corpus-representation covariates. **Human Development Index (HDI)** (UNDP Human Development Report 2023–24, 2022 data) and **GDP per capita** index developmental position and enter the mechanism analysis as adjustment covariates.

The corpus-representation exposures are organized along a distinction that the mechanism analysis (H4) makes explicit, because the two channels apply to different effects. **Language-corpus measures** quantify how much pretraining text exists *in* a language and are the candidate mechanism for the native-language penalty (H2): the mC4 token count per language (12,

Table 6), the OSCAR-2301 deduplicated corpus size per language [13], and the Joshi linguistic-resource class [3]. **Country-corpus measures** quantify how much the encyclopedic corpus is *about* a country, independent of language, and are the candidate mechanism for the geographic gap (H1/H4): the byte length of the English Wikipedia article on the country, the number of Wikipedia language editions with an article on it (Wikidata sitelink count), and the number of Wikidata statements on the country entity. The two families are not interchangeable: because the geographic gap is measured *in English* (language held constant across countries), a residual North/South gap cannot be a language-corpus effect, so H4 isolates the country-representation channel for H1 and the language-corpus channel for H2. All covariate values, official sources, and access dates are versioned in the analysis release.

3.5 Model selection

Fourteen LLMs were benchmarked across five access tiers, spanning roughly two orders of magnitude in parameter count and four training-data geographies (United States, China, Europe, Brazil). **Tier A (open-weight frontier, 5 models)** comprised large open-weight systems accessed via free inference endpoints. **Tier B (open-weight mid, 3 models)** and **Tier C (open-weight small/regional, 2 models)** were run locally via Ollama under pinned model tags; Tier C includes a Brazilian-Portuguese instruction-tuned open model (Cabra-Mistral 7B v3) used for the regional-model test (H3) against a scale-matched globally trained open model. **Tier D (closed accessible, 2 models)** and **Tier E (closed frontier, 1 model)** were accessed via official APIs. Each model was queried with an identical version string or pinned tag throughout the collection window; tags and, where available, weight hashes are recorded in the protocol release to guard against mid-collection model updates. The complete model roster with vendor, parameter count, access route, and execution backend is provided in Supplementary Table S1.

A note on the unit of analysis. We compare *models as served* through a fixed access stack (official vendor API for closed models; a pinned inference endpoint for open-weight models), not idealized model weights in isolation. Provider infrastructure, serving configuration, and API layer are therefore part of what each label denotes, and a residual gap could in principle reflect the stack rather than the weights. We hold this constant by pinning version strings/tags and the maximum-determinism configuration available per provider throughout a fixed collection window, and treat "model" throughout as shorthand for this served-model stack; disentangling weights from infrastructure is a separate question we do not claim to resolve.

3.6 Domain and task taxonomy

The single domain is **air pollution policy**, chosen because it anchors to officially published, verifiable ground truth (national ambient air quality standards, environmental-agency monitoring data, federal control programs) and maps to concrete decision activities of a public environmental manager. Five task types operationalize the domain:

- **T1 — Technical standard.** Recall of binding ambient air quality standards, emission limits, or regulatory thresholds (for example, the annual mean $PM_{2.5}$ standard in force in the country).

- **T2 — Local factual datum.** Recall of officially measured air quality data for a specified city or region and year, as reported by the national or subnational environmental agency.
- **T3 — Health-evidence synthesis.** Synthesis of peer-reviewed epidemiological evidence on the air pollution health burden for the country, scored against a rubric rather than a single value.
- **T4 — Policy instruments.** Identification of policy programs, executing agencies, and legal instruments for air pollution control in force in the country.
- **T5 — Applied recommendation.** Rubric-scored applied recommendation for a defined operational scenario (for example, short-term response to a high-concentration episode under thermal inversion).

T1, T2, and T4 admit verifiable ground truth and carry the primary accuracy signal; T3 and T5 are rubric-scored open-generation tasks. The country-by-task mapping of official source classes, equivalence criteria, and verifiability status is specified in the ground-truth registry (Section 3.9).

3.7 Persona manipulation

Persona is a within-prompt factor with two levels. The **neutral** condition presents the query as a plain information request. The **public_manager_env** condition prepends a fixed role frame identifying the requester as a municipal or regional environmental management official, holding the substantive question identical across conditions. A single role-frame template is applied uniformly across countries and tasks (translated where the prompt language is non-English), so that any between-condition difference is attributable to the persona frame rather than to question wording. Persona order within each prompt-model cell is randomized, and a manipulation check verifies that the role frame is acknowledged in the response. The exact wording, translation procedure, and manipulation-check coding are pre-specified and reproduced in Supplementary Material. Persona is the basis of hypothesis H6, which tests whether the role frame reduces (or amplifies) the country-level accuracy gradient.

3.8 Prompt construction and validation

Prompts were authored for the air pollution policy domain by the research team with domain supervision (PPGSAU/UTFPR) and validated against the ground-truth registry before entering the confirmatory stimulus set. A pilot subset was first validated for inter-rater agreement using Krippendorff's α, with a threshold of $\alpha \geq 0.70$ required before scaling the design. All prompts were authored in English and, for a stratified subset of countries, additionally in a relevant native language (sparse multilingual matrix; Supplementary Table S2); native-language prompts were translated and back-translated by native-speaker translators, and any prompt whose back-translation produced a semantic shift was rewritten or excluded. Following the Brazil pilot validation, every factual claim in a ground-truth entry was required to resolve to an official source URL or a peer-reviewed DOI before the corresponding prompt was admitted, enforcing the project Citation Verification Protocol at the ground-truth generation stage rather than only at the manuscript stage.

3.9 Ground truth and the ground-truth registry

Ground truth for the verifiable tasks (T1, T2, T4) was drawn from official sources of record: national air quality standards and the agencies that issue them (T1), environmental-agency monitoring reports (T2), and federal policy programs with their executing agencies and legal instruments (T4). For T3, the reference is a curated set of peer-reviewed epidemiological studies with resolvable DOIs rather than a single number. Because the validity of the geographic gradient (H1) and the persona effect (H6) depends on ground truth being equally available and equally verifiable across all countries, we constructed a per-country ground-truth registry that, for each (country, task) pair, records the class of equivalent official source, the resolvable URL, the gold value or rubric key, the reference date, the human validator, and a validation flag. The registry design, cross-country equivalence criteria, and the explicit separation of an AI knowledge gap from a ground-truth-availability gap are documented in `docs/ground_truth_registry_design.md` and summarized in Supplementary Material. Entries are versioned by access date, source URL, and a SHA-256 hash of the retrieved document. Where ground truth could plausibly have appeared in training data, a sensitivity subset is constructed from documents dated after each model's training cutoff. Entries whose official source is unavailable or non-equivalent for a country are flagged and excluded from the primary accuracy comparison for the affected (country, task) cell, so that missing ground truth is never scored as a model error.

3.10 Data collection

API and local calls were executed at temperature 0.3 with deterministic seeds where available, with 2 replications per prompt-model-persona combination to estimate stochastic variability within the budget. Collection occurred within a fixed window to minimize model-update confounds, with pinned tags and version strings recorded per call. All responses were stored alongside complete request metadata (model version string or tag, timestamp, token counts, finish reason, persona condition) in a versioned storage layer with SHA-256 checksums. Response quality gates excluded truncated responses ($\texttt{finish\_reason} \neq \texttt{stop}$) and responses in a language divergent from the request; both were retained and analyzed separately as secondary outcomes.

3.11 Measures

Scoring used an LLM-as-judge protocol with a single primary judge and a multi-vendor reliability panel. Every response in the confirmatory set was scored by the **primary judge** (GPT-5-mini) against the ground-truth registry entry (for T1, T2, T4) or the validated rubric (for T3, T5), producing the five normalized subcomponents below. To quantify judge sensitivity, a **fixed stratified reliability sample** (stratified by country × task × persona, fixed random seed) was independently re-scored by a vendor-diverse panel. The operative panel comprises four judges spanning four vendors — GPT-5-mini (OpenAI), Claude Sonnet 4.6 (Anthropic), Gemini 2.5 Pro (Google), and DeepSeek-V3 (DeepSeek) — chosen for vendor diversity so that intra-vendor correlation does not inflate apparent agreement; a fifth judge (Llama 3.3 70B, Meta) corroborates on the subset it covered, and a sixth candidate (Command R+, Cohere) was attempted but excluded when its provider quota was exhausted mid-run. Reliability is reported three ways: Krippendorff's α (an inter-rater agreement coefficient, $1 =$ perfect, $0 =$ chance), the single-judge ICC(2,1) (intraclass correlation for one rater, two-way random effects), and, as the operative quantity, the **ICC(2,k) reliability of the panel mean**, which by the Spearman-Brown relation is high even when individual judges agree only moderately. We report agreement transparently, including the most discordant judge: any systematic leniency offset cancels in the between-group contrasts that carry every reported effect (tier gap, language penalty, task floor), so moderate single-judge agreement does not threaten those contrasts. This study did not include a human gold-standard layer; ground truth was instead anchored to official primary sources through a documentary audit (Section 3.9), and the absence of independent human scoring is stated as a limitation.

The primary composite outcome combines the five pre-specified subcomponents (see Supplementary Table S3), each normalized to $[0, 1]$: (i) factual accuracy against the registry entry (binary T1; partial credit T2–T4); (ii) contextual completeness scored against a pre-validated rubric (T3, T4, 0–5); (iii) citation quality verified via DOI and official-source check; (iv) hallucination absence, coded binarily (0 hallucinated, 1 faithful); and (v) applied validity, the rubric score for the operational recommendation in T5. Primary inference uses equal weights; two pre-specified sensitivity weightings (a PCA-derived set estimated on the BRA extended pilot and an author-specified set: factual 0.30, contextual 0.25, citation 0.15, hallucination 0.15, applied 0.15) are reported, and a conclusion is treated as robust only if it holds across all three. The composite is computed per (country, model, persona) cell to support the H6 difference-in-differences test.

3.12 Analytical strategy

The three primary confirmatory tests are pre-specified as a single family (F1) and corrected together with the Bonferroni-Holm procedure (a step-down family-wise correction for multiple comparisons). Secondary confirmatory tests (F2) are reported unadjusted within family, and declared exploratory tests (F3) use Benjamini-Hochberg FDR at $q = 0.10$ [32].

H1 (geographic gradient, primary). Country-level mean composite accuracy (n=15) is regressed on the Joshi/HDI gradient via Spearman ρ (rank correlation), pre-specified one-sided with threshold $\rho \geq 0.55$ at $\alpha = 0.05$, with a Mann-Kendall test (a non-parametric monotonic-trend test) as a mandatory non-monotonicity complement and a Global South dummy as a sensitivity contrast.

H4 (corpus-representation mechanism, primary). Country-level partial Spearman ρ relates corpus representation to accuracy after adjusting for HDI and log GDP per capita, using Wikipedia counts as the primary proxy ($\rho \geq 0.55$) and requiring convergence across at least two of three Common Crawl operationalizations ($\rho \geq 0.40$). An E-value sensitivity analysis [33] quantifies the unmeasured confounding required to nullify the association (threshold 2.0).

H6 (persona effect, primary). The persona effect is tested as a country-level difference-in-differences on the composite outcome: $\Delta_{\mathrm{GS}} = \overline{\mathrm{persona}}_{\mathrm{GS}} - \overline{\mathrm{neutral}}_{\mathrm{GS}}$ and $\Delta_{\mathrm{GN}} = \overline{\mathrm{persona}}_{\mathrm{GN}} - \overline{\mathrm{neutral}}_{\mathrm{GN}}$, with H6 statistic $\Delta_{\mathrm{GS}} - \Delta_{\mathrm{GN}}$. H6 is supported (one-sided, $\alpha = 0.05$) if the persona condition reduces the Global South gap by at least 0.05 on the $[0, 1]$ composite (equivalently, 5 percentage points of absolute accuracy). Inference uses a country-level permutation test (5,000 permutations) reported alongside a parametric paired test; the two-sided test is reported as a secondary check because persona amplification is theoretically possible and is reported transparently if observed.

H3 (regional model, secondary). A pre-specified one-sided contrast in a generalized linear mixed model compares Cabra-Mistral 7B v3 on Brazilian Portuguese prompts against a scale-matched globally trained open model, with an exploratory companion contrast on other Lusophone contexts to distinguish gap closure from gap displacement.

H2 and H5 (exploratory). The prompt-language $\times$ country interaction (H2) and the open-frontier versus closed-frontier contrast (H5) are reported descriptively, with formal tests under F3 correction where applied.

Cross-cutting mixed-model analyses use Generalized Linear Mixed Models with random intercepts for country and model, implemented via `pymer4` [34] interfacing R's `lme4` [35]: `lmer` with REML for the Gaussian composite and `glmer` with a binomial family and logit link for binary outcomes. Where a mediation decomposition is reported for H4, it uses `semopy` [36] with 5,000 bootstrap replicates. Robustness analyses, all pre-specified, include leave-one-country-out re-estimation, restriction to the human-gold-validated subset, Bayesian re-estimation with weakly informative priors via `pymc` [37], and a separate analysis excluding prompts flagged for potential training-data contamination. A Monte Carlo power simulation (10,000 iterations under pilot-informed effect sizes) confirms family-wise power for the primary trio under Bonferroni-Holm; the within-subject persona design yields a minimum detectable gap reduction of approximately 2–3 percentage points, so the 5-point H6 threshold is well-powered. Power details are reported in Supplementary Material.

3.13 Study protocol and open science

The full protocol — hypotheses, prompts with hashes, the ground-truth registry, the persona template, scoring rubrics, GLMM and DiD specifications, multiple-comparison corrections, and exclusion criteria — is pre-specified in advance of confirmatory data collection. The analytic dataset, after quality gates but before model fitting, is frozen and hashed so that the confirmatory analysis can be audited against the locked specification. The full protocol, prompts, ground-truth references, analysis code, and anonymized response data will be released on publication under CC-BY-4.0 (data and prompts) and the MIT License (code) [38]; a companion Data Paper is prepared for *Scientific Data*.

3.14 Ethical considerations

No personal data from end-users of LLMs were collected. The study analyzes model outputs to publicly relevant policy questions; domain experts who contribute to ground-truth validation are acknowledged as co-authors when they meet ICMJE criteria. LLM provider terms of service were reviewed and respected, and responses are redistributed solely for research under fair-use and reproducibility provisions, excluded from any commercial application.

4 Results

Sample note. We report a pre-registered *15-country* benchmark and its *post-registration extension to 25 countries* (the latter adds six Global North references — the United Kingdom, Canada, Australia, South Korea, France, Italy — and four native-pair countries — Colombia, Chile, Portugal, Angola). Fourteen LLMs are scored across five tasks, two personas, and four languages (English plus a country-native language — Spanish, Portuguese, or Hindi — for the nine applicable countries, giving the within-country English/native pairs that test H2). Every effect computed on the 25-country sample is reported alongside its pre-registered 15-country value; no pre-registered conclusion is revised on the extension alone. All responses are scored by GPT-5-mini, validated against a four-vendor judge panel (Section 4.11); the T1 ground truth is anchored to official primary sources for all 25 countries (Section 4.9).

4.1 Sample and judging

The analysed corpus comprises 9,251 judge-scored responses on the $[0, 1]$ composite, of which 7,580 are English-prompt responses across the 25 countries (the remainder are the within-country native-language pairs used for H2). The primary judge is GPT-5-mini. Because GPT-5-mini is itself among the evaluated tier, three additional vendor-diverse judges re-scored a fixed stratified sample to quantify judge sensitivity; agreement is strong (panel-mean reliability $\mathrm{ICC}(2, 4) = 0.89$; Section 4.11), which neutralizes the judge–target overlap for the reported composites.

4.2 Accuracy by model

Table 1 orders the 14 models by mean composite. Frontier systems (GPT-5, GPT-5-mini, DeepSeek-V3) lead; small open-weight models trail. The result bearing on H3 is unambiguous: the Brazilian-Portuguese regional model Cabra-Mistral 7B is the *weakest* of all models (0.320 versus 0.543 for the rest; Mann-Whitney $p \approx 0$, Cliff's $\delta = -0.51$, a non-parametric effect size in $[-1, 1]$), contradicting the optimistic H3 framing that a regional model buys domain accuracy. Scale and frontier training dominate regional instruction tuning at the 7B level.

Table 1: Confirmatory composite accuracy by model across 25 countries (GPT-5-mini judge, $[0, 1]$). N = scored English-prompt responses.

Model	N	Mean
GPT-5	780	0.647
GPT-5-mini	777	0.626
DeepSeek-V3	500	0.625
Gemini 2.5 Flash	490	0.604
Claude Haiku 4.5	490	0.597
Qwen3 32B	498	0.546
Command R+	500	0.509
Llama 3.3 70B	474	0.506
GPT-OSS 120B	298	0.485
Phi-4 14B	500	0.485
Qwen3 14B	500	0.474
Llama 4 Scout	500	0.473
Llama 3.1 8B	777	0.424
Cabra-Mistral 7B	496	0.320

4.3 Accuracy by task: the factual-recall floor

Difficulty is sharply task-dependent (Table 2). The two hardest tasks are **T1 (technical-standard recall, 0.367)** and **T2 (local measured datum, 0.368)** — precisely the tasks that require retrieving a specific binding value (a national annual $PM_{2.5}$ standard; a city-year concentration). The open recommendation task (T5) scores highest (0.773) because its rubric rewards plausible, well-structured policy advice that does not hinge on a single fact. The two factual-recall tasks (T1, T2; pooled mean 0.367) score far below the synthesis and recommendation tasks (T3–T5, 0.638); the difference is large and unambiguous (Mann-Whitney $p \approx 0$, Cliff's $\delta = -0.64$). This is the most robust finding in the study: LLMs are weakest exactly where applied environmental management needs them most — recalling the precise regulatory threshold or measured value in force.

Table 2: Confirmatory composite accuracy by task type (25 countries).

Task	N	Mean
T5 (applied recommendation)	1,508	0.773
T4 (policy instruments)	1,494	0.574
T3 (health-evidence synth.)	1,519	0.567
T2 (local factual datum)	1,530	0.368
T1 (technical standard)	1,529	0.367

4.4 H1 — a tier gap, and (at $n = 25$) a development gradient

H1 was pre-specified in two parts. The **tier gap** holds in both samples: the ten Global North countries average 0.567 against 0.505 for the fifteen Global South countries, a gap of **+6.2 percentage points** whose 95% bootstrap confidence interval over countries is $[+3.7, +8.6]$ pp and *excludes zero* (15-country: +6.7 pp, $[+2.8, +10.2]$). The **monotonic gradient** behaves differently across samples, and this is where the extended sample changes the picture. At the pre-registered $n = 15$ the gradient was undetectable (Spearman $\rho(\text{accuracy}, \text{HDI}) = +0.14$, $p = 0.31$; Mann-Kendall $p = 0.69$). In the extended $n = 25$ sample it becomes significant: $\rho(\text{accuracy}, \text{HDI}) = +\mathbf{0.51}$ ($p = 0.004$, one-sided), with a significant Mann-Kendall trend ($S = 102$, $Z = +2.36$, $p = 0.018$). The development gradient is thus real but was *underpowered at* $n = 15$; the seven added high-HDI references (Table 3: South Korea, Italy, France, Canada, Australia, the UK, Portugal all cluster near the top) increase the density of high-development observations enough to reveal it — without widening the HDI range itself (Section 4.12).

Two honest qualifications bound this result. First, $\rho = 0.51$ *falls just short of our pre-registered* $\rho \geq 0.55$ *effect-size threshold*: the gradient is statistically significant but below the magnitude we pre-committed to call a strong gradient. Second, the gradient is in *development* (HDI), *not* in linguistic resource: the Joshi-class correlation remains null ($\rho = -0.06$). The country ordering also retains large axis-orthogonal heterogeneity — India (0.631, Global South) still tops all countries and Germany (0.535) is the lowest-scoring Global North country — so the disadvantage is a development gradient layered over substantial country-specific variation, not a clean ranking. Under Bonferroni-Holm across the primary family {H1-gradient, H4, H6}, H1-by-HDI is the one test that rejects its null at $n = 25$ ($p = 0.004$ versus threshold 0.0167).

Table 3: Confirmatory composite accuracy by country, 25-country sample (tier: GN = Global North, GS = Global South). Ordered by mean.

Country	Tier	Mean	Country	Tier	Mean
India	GS	0.631	Germany	GN	0.535
South Korea	GN	0.601	Chile	GS	0.528
Japan	GN	0.588	South Africa	GS	0.522
Italy	GN	0.584	Angola	GS	0.506
United States	GN	0.581	Bangladesh	GS	0.503
France	GN	0.564	Philippines	GS	0.499
Canada	GN	0.562	Mexico	GS	0.496
Australia	GN	0.559	Peru	GS	0.496
United Kingdom	GN	0.554	Kenya	GS	0.488
Portugal	GN	0.550	Nigeria	GS	0.465
Indonesia	GS	0.540	Egypt	GS	0.465
Colombia	GS	0.539	Argentina	GS	0.450
			Brazil	GS	0.449

4.5 H4 — corpus representation: country coverage, not language size

The pre-specified mechanism for the geographic gap was training-corpus representation, with the pre-registered proxy being Wikipedia article/edition count. That proxy is **null** at both $n = 15$ and $n = 25$ (Spearman $\rho = +0.08$; partial controlling HDI $+0.04$, $p = 0.87$): **H4 as pre-registered is not supported**. We then explored, post hoc, whether a different channel is at work. Because the gap is measured *in English*, a residual North/South difference cannot reflect how much text exists in a country's language, so we distinguish language-corpus size from how broadly the corpus is *about* a country. Among three country-coverage proxies, the number of Wikipedia language editions with an article on the country (Wikidata sitelinks) correlates with accuracy at the zero-order level ($\rho = +0.54$, $p = 0.005$, which survives Benjamini-Hochberg correction across the three proxies, adjusted $p = 0.016$; Wikidata statements $\rho = +0.31$ and English-article byte length $\rho = +0.20$ are both non-significant). *However*, this association **attenuates to non-significance after adjusting for HDI** (partial $\rho = +0.32$, $p = 0.13$), so it does not meet the partial-correlation standard the mechanism test requires. We therefore report country-coverage breadth as a *suggestive, exploratory* signal — consistent with a gap that appears even when every country is queried in English, and a more promising lead than language-corpus size — but **not** as an established mechanism: the development gradient and country coverage are entangled, and the present data cannot separate them.

For the native-language penalty (H2), the relevant channel is instead language-corpus size, and the direction is as predicted, though with only three native languages it is descriptive: the per-language penalty falls monotonically with mC4 token volume [12] and OSCAR size [13] (Spearman $= -1.00$ over the three) — Hindi, the smallest corpus (mC4 24B tokens), carries the largest penalty, and Spanish (433B) the smallest.

4.6 H6 — persona does not narrow the geographic gap

The persona factor was fully crossed, so H6 is tested as a difference-in-differences. The public-environmental-manager frame leaves the North/South gap essentially unchanged: +6.4 pp under the neutral frame versus +6.0 pp under the manager frame, a DiD of +**0.4** pp, far below the pre-specified 5 pp threshold; a country-level permutation test (5,000 permutations of the tier label) gives $p = 0.26$. **H6 is not supported** in either sample (15-country DiD +1.0 pp, $p = 0.14$): presenting the query as coming from a public environmental manager neither narrows nor amplifies the Global South disadvantage, consistent with evidence that expert personas do not reliably improve factual accuracy [27, 28]. This answers the study's central applied question in the negative — prompt framing is not a fix for geographic bias.

4.7 H5 — open versus closed-accessible

Grouping by access type, closed-accessible models average 0.623 ($n = 2{,}537$) against 0.481 for open-weight models ($n = 5{,}043$), a +14.1 pp closed advantage. This runs against the optimistic H5 framing; the open arm includes larger models (Llama 3.3 70B, DeepSeek-V3), so the gap is not attributable to scale alone, although DeepSeek-V3 itself ranks third overall. H5 is interpreted descriptively.

4.8 H2 — native-language prompting lowers accuracy

For the nine countries with a target native language (Brazil/Portugal/Angola → Portuguese, Mexico/Argentina/Peru/Colombia/Chile → Spanish, India → Hindi), each prompt is rendered in both English and the native language, holding content fixed, so the English/native contrast is within-country and within-model. Across the matched (model, prompt) cells, native-language prompting *lowers* mean composite accuracy by 2.1 percentage points (Wilcoxon signed-rank $p = 2 \times 10^{-3}$, $n = 762$ cells). Because those cells are clustered within models and prompts, we confirm the effect at the *country* level, where the unit is independent: aggregating to one English-minus-native delta per country, all nine native-pair countries show a penalty (9/9 negative, mean −2.4 pp, Wilcoxon $p = 0.008$), so it is not an artifact of treating clustered cells as independent. The effect is modulated by the resource level of the native language (Table 4): the Spanish penalty is small and non-significant (−1.0 pp, $p = 0.15$), the Portuguese penalty is now significant with the added Portugal/Angola pairs (−2.3 pp, $p = 0.033$), and the Hindi penalty is large (−7.8 pp, $p = 0.013$). India, which scores highest of all countries in English, drops sharply in Hindi. The applied implication mirrors H6: the intuitive fix — letting a local manager ask in their own language — does not help and, for a non-Latin-script lower-resource language, materially hurts. This is consistent with the linguistic-resource hierarchy [3] and with the language-corpus channel of H4.

Table 4: H2: native-minus-English accuracy by native language (paired within country and model, 25-country sample).

Native language (Joshi class)	Countries	Δ (pp)	N pairs
Spanish (5)	MEX, ARG, PER, COL, CHL	−1.0	412
Portuguese (4)	BRA, PRT, AGO	−2.3	284
Hindi (4)	IND	−7.8	66
Overall		−2.1	762

4.9 Ground-truth provenance

Because measured "accuracy" depends on the ground truth, the technical-standard target (T1) for all 25 countries was anchored to official primary sources by documentary verification rather than expert assertion, and is reproducible from the cited URLs. Most countries have a value read from the official regulation (e.g. Brazil's annual $PM_{2.5}$ standard of 17 μg/m^3 read verbatim from Annex I of CONAMA Resolution 506/2024 in the Diário Oficial da União; Colombia's 25 μg/m^3 read from Tabla 1 of Resolución 2254/2017). Three countries have *no* national $PM_{2.5}$ standard, confirmed from the official text — Nigeria's NESREA regulation sets PM10 only, Argentina has no binding federal value, and Angola has no national air-quality legislation — a fact the audit established against circulating but incorrect secondary figures, and in which a faithful model must decline to state a non-existent threshold rather than fabricate one. This provenance is more reproducible than expert validation: any reader can re-verify each value from the official source.

4.10 Qualitative failure modes

To make the composite numbers concrete, Box 1 shows three representative failures drawn verbatim from the response logs (run IDs available in the release).

Table 5: Box 1 — Representative failure modes (verbatim, lightly truncated).

Failure mode	Example
Fabricating a non-existent standard (country with no national $PM_{2.5}$ norm)	For **Angola** (no national standard), Gemini 2.5 Flash asserted a "25 μg/m^3" annual standard in one replicate and "15 μg/m^3" in another; GPT-5 asserted "15 μg/m^3" for **Nigeria**, which regulates only PM_{10} (NESREA). A faithful response would decline to state a value that does not exist.
Missing the binding value (T1 floor)	For **Brazil** (official annual standard 17 μg/m^3, CONAMA 506/2024), the modal model answer was 15 μg/m^3 (42 responses), with 10, 20, and 25 also common; the official value was almost never returned.
Native-language degradation (H2)	For **India** (annual standard 40 μg/m^3), GPT-OSS 120B returned the correct value in English but an *empty* answer to the same question in Hindi.

The first mode is the most consequential for the no-standard cases (Angola, Nigeria, Argentina): rather than abstaining, models invent a plausible regulatory cutoff and disagree across replicates and models on the invented value — exactly the behaviour a public manager cannot detect without the official source.

4.11 Inter-judge reliability

To address the judge–target overlap (GPT-5-mini is in the evaluated set), a fixed stratified reliability sample of 131 responses was independently re-scored by a vendor-diverse judge panel: GPT-5-mini (OpenAI), Claude Sonnet 4.6 (Anthropic), Gemini 2.5 Pro (Google), and DeepSeek-V3 (DeepSeek). Reliability is reported three ways: Krippendorff's $\alpha = 0.667$ (interval), the single-judge $\text{ICC}(2,1) = 0.672$, and — the operative quantity — the reliability of the panel *mean*, ICC$(2, 4) = \mathbf{0.891}$, which by the Spearman-Brown relation is high even though individual judges agree only moderately. Pairwise agreement is strongest between GPT-5-mini, Claude, and DeepSeek (Pearson 0.82–0.86) and lowest for the systematically more lenient Gemini (0.61–0.63); because all reported effects are between-group contrasts, this leniency offset cancels. A fifth judge (Llama 3.3 70B, Meta) corroborated on the subset it covered (pairwise Pearson 0.51–0.60; five-judge ICC$(2, 5) = 0.86$) but, being a noisier 70B judge, slightly *lowered* the panel α — empirically confirming that four strong judges are preferable to five with a weaker one. A sixth candidate (Command R+) was excluded when its provider quota was exhausted mid-run. The study uses the single GPT-5-mini composite for all effects, validated by this panel, rather than a judge-averaged score.

4.12 Robustness of the development gradient

Because the gradient is significant only in the extended sample, we test two threats directly. **Range extension.** The concern is that adding ten countries — seven of them Global North — manufactures the gradient by stretching the HDI predictor. It does not: the ten added countries fall *within* the HDI range already spanned by the original fifteen ($[0.548, 0.950]$, from Nigeria to Germany), so the extension increases the *density* of observations, not the range, and the Spearman estimate is identical on the range-restricted set ($\rho = 0.512$, $p = 0.009$). The gradient strengthened from $n = 15$ to $n = 25$ not because the predictor range widened but because the added high-HDI references score consistently well, sharpening an association the smaller sample was underpowered to detect. **Influential observations.** Leave-one-country-out re-estimation gives an HDI-gradient range of $\rho \in [+0.48, +0.66]$ across all 25 drops, so no single country drives it; notably, dropping India — the Global South country that scores highest and is the strongest counterexample to the gradient — *raises* the correlation to $\rho = 0.66$. India is therefore not propping up the gradient but attenuating it, and the same leave-one-out exercise leaves the tier gap ($[+5.8, +7.1]$ pp) and the exploratory sitelinks correlation ($[+0.50, +0.63]$) stable. We read India as consistent with — not contrary to — the country-coverage account: it is among the most extensively documented countries in encyclopedic and multilingual sources, exactly the condition under which the coverage signal predicts higher accuracy, which is why it scores highest in English despite its development level.

Metric robustness. The composite blends five subcomponents, so one might worry it dilutes or manufactures the effects. Restricting to the single most objective subcomponent — factual accuracy against the registry value, ignoring completeness, calibration, and style — *strengthens* every headline result: the tier gap widens to +10.7 pp (versus +6.2 on the composite), and the factual-recall floor deepens (T1+T2 factual accuracy 0.25 versus 0.71 for T3–T5). The composite is therefore conservative; the geographic and task effects are not artifacts of mixing subcomponents.

4.13 Summary

Table 6 maps each hypothesis to its finding, the candidate mechanism, and an explicit evidence status, so that no claim is read as stronger than the data support.

Table 6: Hypotheses, findings, candidate mechanisms, and evidence status.

Hypothesis	Finding	Candidate mechanism	Evidence status
H1 (tier gap)	+6.2 pp, CI excludes zero	development (tier)	Supported
H1 (gradient)	$\rho = 0.51$ with HDI ($n = 25$)	development (HDI)	Supported; below pre-registered 0.55
H2 (language)	−2.1 pp; Hindi −7.8	language-corpus size	Effect supported; mechanism descriptive ($n = 3$)
H3 (regional model)	$\delta = -0.51$	scale > regional tuning	Supported
H4 (geographic mech.)	sitelinks $\rho = 0.54$	country coverage	Exploratory (attenuates with HDI)
H4 (pre-registered)	Wikipedia-size null	language-corpus size	Not supported
H5 (open/closed)	+14.1 pp closed	access type / scale	Descriptive
H6 (persona)	DiD +0.4 pp	—	Not supported

The expansion sharpens the central result while leaving the secondary findings intact. **Supported, with effect sizes:** a Global North/South *tier* gap (+6.2 pp, bootstrap 95% CI $[+3.7, +8.6]$, excludes zero) and, at $n = 25$, a significant *development gradient* (Spearman $\rho = 0.51$ with HDI, $p = 0.004$; Mann-Kendall $p = 0.018$; rejects under Bonferroni-Holm) that was underpowered at the pre-registered $n = 15$; a native-language *penalty* (overall $p = 2 \times 10^{-3}$, Hindi −7.8 pp, Portuguese now significant); a technical-/local-recall *floor* (T1+T2 vs others, $\delta = -0.64$); and the regional model as the *worst* performer ($\delta = -0.51$). **Not supported (informative nulls):** the gradient is *not* linguistic-resource (Joshi $\rho = -0.06$) and falls just short of the pre-registered 0.55 effect-size threshold; the pre-registered corpus proxy (Wikipedia size) is *not* the mechanism (partial $\rho = 0.04$), and a country-coverage proxy (sitelinks $\rho = 0.54$, $p = 0.005$) is suggestive but *not* robust to HDI adjustment (partial $p = 0.13$), leaving the mechanism exploratory; and persona does *not* narrow the gap (permutation $p = 0.26$). The geographic disadvantage the geographic disadvantage is a real, leave-one-out-robust development gradient whose mechanism we localize only *suggestively* to country representation rather than language-corpus size; the dominant robust harms are *linguistic* (native-language prompting, worst for Hindi) and *task-specific* (factual recall of the regulation or datum in force); and all three intuitive remedies — a local-manager persona, the local language, and a regional model — fail. Ground truth is anchored to official primary sources for all 25 countries, and panel-mean reliability is strong ($\text{ICC}(2, 4) = 0.89$). This matters for environmental management in the Global South, where the disease burden is large [1, 39] and where exactly the hardest tasks — recalling the binding standard or measured value — are the ones a manager most needs [3, 4].

5 Discussion

We set out to measure whether LLMs answer air pollution policy questions as reliably for the Global South as for the Global North, whether prompt framing changes the answer, and what mechanism drives any gap. The benchmark — 25 countries, 14 models, four languages, five tasks, two personas, scored by a panel-validated judge against official ground truth — returns a disciplined and partly counter-intuitive picture. A geographic disadvantage exists and, with adequate power, resolves into a genuine development gradient; but its mechanism is country representation rather than language-corpus size, the dominant harms are linguistic and task-specific, and every intuitive remedy fails. We organise the discussion around the gradient and its mechanism, the robust harms, the three failed remedies, the public-health stakes, the limitations that bound the claims, and what follows for practice.

5.1 The geographic gap is a development gradient — visible only at power

At the pre-registered $n = 15$, the North/South *tier* gap was real (+6.7 pp, CI excludes zero) but the *gradient* was undetectable (Spearman vs HDI $\rho = 0.14$, $p = 0.31$). Extending the sample to 25 countries — adding seven high-HDI Global North references that thicken the high-development end of the sample (without widening its range; Section 4.12) — the gradient resolves into significance ($\rho = 0.51$ with HDI, $p = 0.004$; Mann-Kendall $p = 0.018$), the one primary-family test that rejects its null under Bonferroni-Holm. We are deliberately transparent that this is a post-registration extension and that $\rho = 0.51$ falls just short of the pre-specified 0.55 effect-size threshold: the honest statement is that the gradient is significant but of moderate magnitude, and that it was underpowered, not absent, in the registered design. This is itself a methodological lesson for the field — geographic-bias gradients across a handful of countries are easy to miss at small n, and the $n = 15$ null would have been the wrong conclusion.

Crucially, the gradient is in *development* (HDI), not in linguistic-resource class (Joshi $\rho = -0.06$), and it coexists with large country-specific heterogeneity: India tops all 25 countries despite being Global South, and Germany is the lowest-scoring Global North country. The disadvantage is a development gradient layered over substantial idiosyncrasy, not a clean ranking, which is why the tier contrast and the gradient are complementary rather than redundant descriptions.

5.2 A suggestive mechanism: country coverage over language-corpus size

Our pre-registered mechanism test fails: the Wikipedia-size proxy is null (partial $\rho = 0.04$), so H4 as registered is not supported. Exploring further, we find a more promising lead. Because the gap is measured *in English*, with the query language held constant, a residual North/South difference cannot be an effect of how much text exists in a country's language; and the candidate that *does* track accuracy is how broadly the corpus represents the country — Wikidata sitelinks correlate at $\rho = 0.54$ ($p = 0.005$, surviving multiplicity correction across three coverage proxies). We are deliberately cautious about this result: the association *attenuates to non-significance after adjusting for development* (partial $\rho = 0.32$, $p = 0.13$), so country coverage and HDI are entangled and the present design cannot establish that coverage *causes* the gap. We therefore offer the country-coverage-over-language-size contrast as a *suggestive, testable refinement* of the "less data for under-represented regions" intuition — one that the India case illustrates (its very high multilingual encyclopedic coverage accompanies its top English score) and that the leave-one-out analysis shows is not driven by any single country — rather than as an established mechanism. If it replicates under a design that can separate coverage from development, it would reframe mitigation toward how comprehensively a country is described in widely-mirrored sources rather than the size of its language's web corpus.

5.3 The robust harms are linguistic and task-specific

Two findings are large, significant, and stable across both samples. First, the **factual-recall floor**: the two tasks that require a specific binding value — the national standard (T1) and a measured city-year concentration (T2) — score around 0.37, far below synthesis and recommendation (0.64; $\delta = -0.64$). The floor sits precisely on the binding values a manager most needs to trust. Second, the **native-language penalty**: asking in the local language lowers accuracy (-2.1 pp overall, $p = 0.002$), trivially for high-resource Spanish but materially for Hindi (-7.8 pp). The penalty tracks language-corpus size across the three native languages, the mechanism the resource hierarchy predicts [3, 12]. Together these say the practical risk is concentrated, not diffuse: a manager retrieving a regulatory cutoff, especially in a lower-resource language, is in the worst cell of the design.

5.4 Three intuitive remedies, all failing

The applied contribution is largely negative. **The local-manager persona** does not narrow the gap (DiD +0.4 pp, permutation $p = 0.26$); presenting the query as coming from a public environmental manager changes nothing, consistent with evidence that expert personas do not improve factual accuracy [27, 28]. **The local language** does not help and can hurt (H2). **The regional model** — Brazilian-Portuguese Cabra-Mistral 7B — is the *worst* performer of all 14 ($\delta = -0.51$), so reaching for a locally tuned small model trades away the scale and frontier training that actually carry domain accuracy. Practitioners routinely deploy all three of these as fixes; none survives contact with the data. The honest message for Global South environmental agencies is that none of the cheap prompt- or model-level workarounds substitutes for verifying the binding value against the official source.

5.5 Why the domain matters

The stakes are not benchmarking abstractions. Fine particulate matter is among the leading environmental risk factors for premature mortality, and the burden falls disproportionately on the Global South; the Global Burden of Disease attributes on the order of fifty thousand deaths per year in Brazil alone to $PM_{2.5}$ exposure [1], and short-term exposure continues to drive measurable mortality in Brazilian cities [39]. When a manager delegates a normative lookup or an episode recommendation to an LLM, an inaccurate answer can propagate into a regulatory brief or an emergency response. The alignment is the worrying part: the regions in the lower classes of the resource hierarchy [3] and with thinner encyclopedic coverage are also where air pollution kills the most people, so the model is least reliable exactly where the consequences are greatest. We read the whole pattern within the coloniality-of-knowledge framework [4]: LLMs are compressed summaries of corpora produced under uneven digital and publication access, and the development gradient, the country-coverage mechanism, and the language penalty are three measurable links from corpus asymmetry to knowledge asymmetry.

5.6 Limitations

Single primary judge. All effects use the GPT-5-mini composite. We address the judge–target overlap with a four-vendor panel on a fixed sample (panel-mean $\mathrm{ICC}(2,4) = 0.89$), but we do not average judges for the primary score, and individual-judge agreement is only moderate ($\alpha = 0.67$); a more lenient fifth judge lowered α, underscoring residual judge sensitivity. Because all reported effects are between-group contrasts, a judge's leniency offset cancels, which is what makes the single-judge composite defensible here.

No human gold layer. Ground truth is anchored to official primary sources by documentary audit, not independent human scoring. This is highly reproducible (every value re-verifiable from a cited URL) but does not capture expert judgement on the rubric-scored tasks (T3, T5). Two features partly mitigate the concern that the scores reflect the judge rather than the responses: the four-vendor judge panel ($\mathrm{ICC}(2,4) = 0.89$), and the fact that restricting to objective factual accuracy strengthens rather than weakens the effects (Section 4.12), so the headline gaps are not an artifact of a permissive composite. A blinded human-expert validation of a stratified response sample, scored against the same rubric with reported human-judge agreement, remains the natural next step and is deferred to the co-author panel.

Post-registration expansion. The 25-country result is an extension decided after the registered analysis; we report it alongside the $n = 15$ values and treat it as a power-and-robustness check, but a confirmatory replication at $n = 25$ would strengthen the gradient claim, which currently sits just under the pre-specified effect-size threshold.

Coarse covariates and three native languages. The corpus proxies are public and reproducible but coarse; the language-corpus mechanism for H2 rests on only three native languages and is reported descriptively; and Oceania-native and Joshi Class 0 languages remain out of scope.

Single domain. All effects come from one applied domain (air pollution policy) and one family of corpus proxies (Wikimedia). The development gradient and the country-coverage signal may not transfer to other policy domains with different documentation patterns, and we confine the mechanistic and "tractable-lever" language to this domain rather than to LLM geographic bias in general; cross-domain replication is the natural next step.

5.7 Implications

For environmental agencies in the Global South the operational guidance is concrete: treat LLM outputs on binding standards and measured data as drafts to be checked against the official gazette, do not assume a local-language query or a regionally tuned small model improves accuracy, and do not rely on role framing to close the gap. For model developers, the country-coverage result points to a tractable lever — breadth of encyclopedic representation — distinct from raw language-corpus volume. For the measurement community, the $n = 15$-to-$n = 25$ reversal is a caution that geographic-bias gradients are easily underpowered. These are the links the study set out to probe, and the data turn each from a plausible worry into a measured effect with an official-source-anchored ground truth.

6 Conclusion

Large language models are becoming working infrastructure for air pollution policy, a domain where an inaccurate answer about a binding standard, a measured concentration, or an operational response carries direct public-health stakes in the Global South, where the $PM_{2.5}$ mortality burden is heaviest [1, 39]. We measured geographic and persona-modulated bias in that domain with a 25-country, 14-model, four-language benchmark, scored by a panel-validated judge against ground truth anchored to official primary sources for every country.

The geographic disadvantage is real and, given adequate power, resolves into a significant development gradient (Spearman $\rho = 0.51$ with HDI at $n = 25$, having been undetectable at the pre-registered $n = 15$, robust to leave-one-out but below our pre-registered effect-size threshold), alongside a stable +6.2 pp North/South tier gap. The mechanism is harder to pin down: the pre-registered corpus proxy is null, and a country-coverage proxy correlates with accuracy ($\rho = 0.54$) but does not survive adjustment for development, so we report country representation — not language-corpus size — as a *suggestive* lead coherent with a gap measured in English, not as an established mechanism. The dominant harms are linguistic (native-language prompting lowers accuracy, worst for Hindi) and task-specific (a sharp factual-recall floor on standards and measured data, $\delta = -0.64$). And the three intuitive remedies all fail: a local-manager persona does not narrow the gap ($p = 0.26$), the local language does not help and can hurt, and the Brazilian-Portuguese regional model is the weakest of all systems ($\delta = -0.51$).

The honest message for Global South environmental agencies is therefore concrete and largely cautionary: LLM outputs on binding standards and measured data should be treated as drafts checked against the official gazette, and none of the cheap prompt- or model-level workarounds — role framing, the local language, a regionally tuned small model — substitutes for that verification. For model developers, the country-coverage result identifies a tractable lever distinct from raw corpus volume. We read the pattern within the coloniality-of-knowledge framework [4]: the regions under-served in training corpora [3] are the same ones where air pollution kills the most people, and this study turns two of those links — a development gradient and a language penalty — from plausible worries into measured effects, while localizing a third (country coverage) only suggestively. We also report a methodological caution: the $n = 15$-to-$n = 25$ reversal shows geographic-bias gradients are easily underpowered, and we flag the expansion as a post-registration extension whose gradient sits just under our pre-specified effect-size threshold. Protocol, prompts, official-source ground-truth registry, analysis code, and anonymized response data are released openly.

Data and code availability

The full protocol, prompts, and analysis code will be released on publication at github.com/Roverlucas/artigo-vies-analise-regional under MIT (code) and CC-BY-4.0 (data/prompts/documentation) licenses, together with the ground-truth registry and anonymised LLM responses. The analysis pipeline is reproducible end-to-end via `python code/run_all.py`.

Study protocol

The statistical analysis plan was pre-specified for the 15-country confirmatory design. Confirmatory analyses follow those specifications; the post-registration extension to 25 countries and the exploratory analyses (notably the H4 country-coverage proxy) are labeled explicitly and reported alongside their pre-registered 15-country counterparts. The full protocol, prompts, ground-truth registry, and analysis code will be released on publication.

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Author contributions

Following CRediT taxonomy:

- **Lucas Rover (UTFPR):** Conceptualization, Methodology, Software, Data Curation, Formal Analysis, Investigation, Visualization, Writing – Original Draft, Project Administration.
- **Dr. Eduardo Tadeu Bacalhau (UFPR):** Validation, Supervision, Writing – Review & Editing.
- **Dra. Yara de Souza Tadano (UTFPR):** Validation, Supervision, Writing – Review & Editing, Funding Acquisition.

Validation, supervision, and final manuscript review are shared by Dr. Eduardo Tadeu Bacalhau (UFPR) and Dra. Yara de Souza Tadano (UTFPR).

Declaration of interests

The authors declare no competing interests.

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